Cloud Computing

CAT7002 Course Notes

Servers, virtualization, data centres and the addressing that connects them.

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Contents

Unit 1: Virtualization Concepts and Hypervisors

1.1. Introduction to Servers

Short description: A server is a computer or software system that provides services, resources, or data to other computers, called clients, over a network.

Everyday examples:

- Opening a website: the browser sends a request to a **web server**.
- Checking Gmail: a **mail server** delivers the messages.
- Using online banking: a **database server** stores and retrieves the account.

What a server does: Stores files, hosts websites, manages databases, runs applications, sends and receives email, shares printers and other resources, and manages users and security.

1.1.1 Types of servers

Server type	Purpose
Web server	Hosts websites
File server	Stores and shares files
Database server	Stores databases
Mail server	Sends and receives emails
Application server	Runs business applications
Print server	Manages printers
DNS server	Converts domain names into IP addresses

1.2. The Traditional Server Model

Short description: Before virtualization and cloud computing, organizations typically deployed **one physical server for one application**. Each application had its own dedicated hardware.

Application	Physical server
Website	Server 1
Database	Server 2
Email	Server 3
ERP	Server 4

1.2.1 Advantages

- **High performance.** Applications have direct access to hardware resources.
- **Better security.** Each application runs on its own physical machine.
- **Reliability.** No interference from other applications sharing the hardware.
- **Simple architecture.** Easy to install, configure and troubleshoot.
- **Full hardware control.** Administrators control the server hardware completely.

1.2.2 Disadvantages

- **Low resource utilization.** Most servers use only 10–20% of CPU and memory, leaving the rest idle.
- **High cost.** Each application requires separate hardware.
- **High power consumption.** More servers draw more electricity and need more cooling.
- **More physical space.** Multiple servers occupy valuable rack space.
- **Difficult scalability.** A new application often means buying new hardware.
- **Higher maintenance.** Every server needs updates, monitoring, backups and repairs.
- **Single point of failure.** If a physical server fails, its application becomes unavailable.

Why a new technology was needed

As organizations expanded they faced too many physical servers, rising hardware costs, increased electricity and cooling expenses, underutilized resources, complex management, and slow deployment of new services. These limitations led to **virtualization**, which allows multiple virtual servers to run on a single physical machine, and virtualization in turn became the foundation for **cloud computing**.

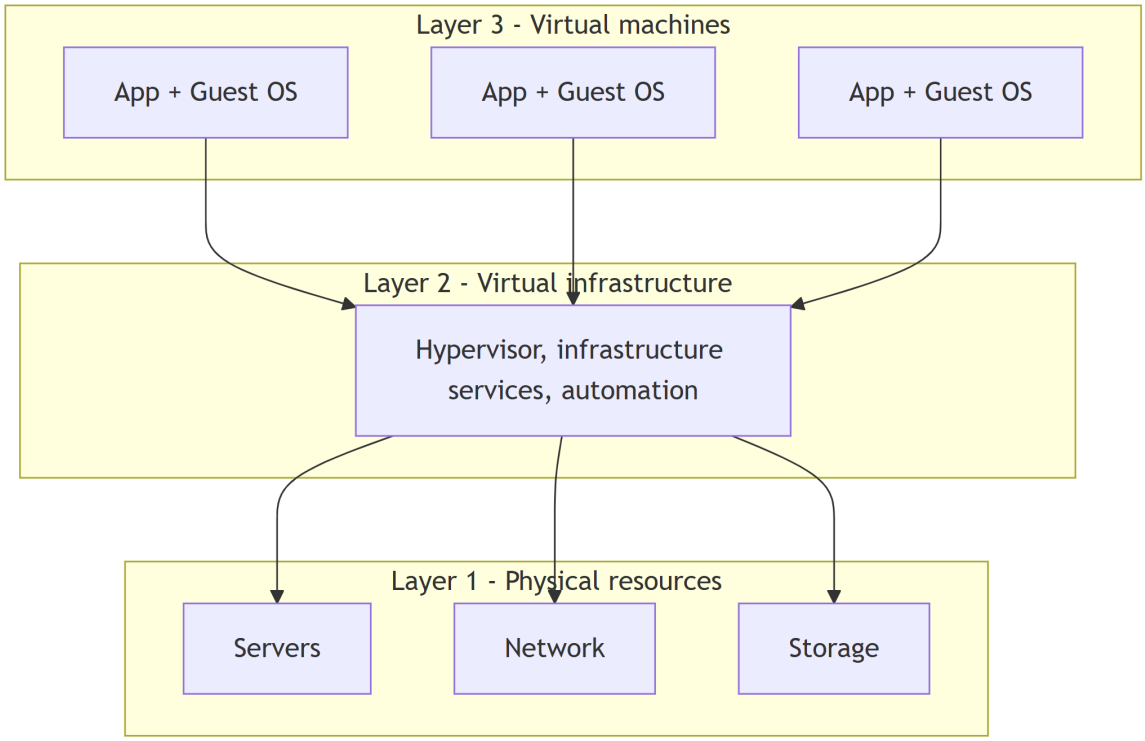
1.3. Introduction to Virtualization

Short description: Virtualization is a methodology for dividing computer resources into more than one execution environment, applying concepts such as partitioning, time-sharing, machine simulation and emulation.

Theory: Virtualization is a method in which multiple independent operating systems run on one physical computer. It maximizes the usage of available physical resources, so one can achieve high server usage: it increases total computing power while decreasing overhead. Increasing the number of logical operating systems on a single host reduces hardware acquisition and maintenance cost for an organization.

How it works: The architecture has three layers. Layer 1 is the physical infrastructure of servers, networks and storage. Layer 2 is the virtual infrastructure, consisting of hypervisors, virtual infrastructure services and automated solutions that optimize the IT process. Layer 3 holds the virtual machines, where different operating systems and applications are deployed. A single virtual infrastructure supports more than one virtual machine, so the physical resources of the whole infrastructure are shared across many virtual machines for maximum efficiency.

1.3.1 Architecture diagram



Notes

- Virtualization is a trend in IT that includes autonomic computing and utility computing: the environment manages itself, and every resource is seen as a utility that a client pays per use.
- It reduces the burden of workloads on users by centralizing administrative tasks and improving scalability.

1.3.2 Reasons for using virtualization

- Virtual machines consolidate the workloads of under-utilized servers, saving on hardware, environmental costs and management.
- Legacy applications can keep running inside a VM.
- A VM provides a secure sandbox for running an untrusted application.
- A VM helps in building a secured computing platform.
- A VM provides an illusion of hardware.
- A VM simulates networks of independent computers.
- A VM supports running distinct operating systems with different versions.

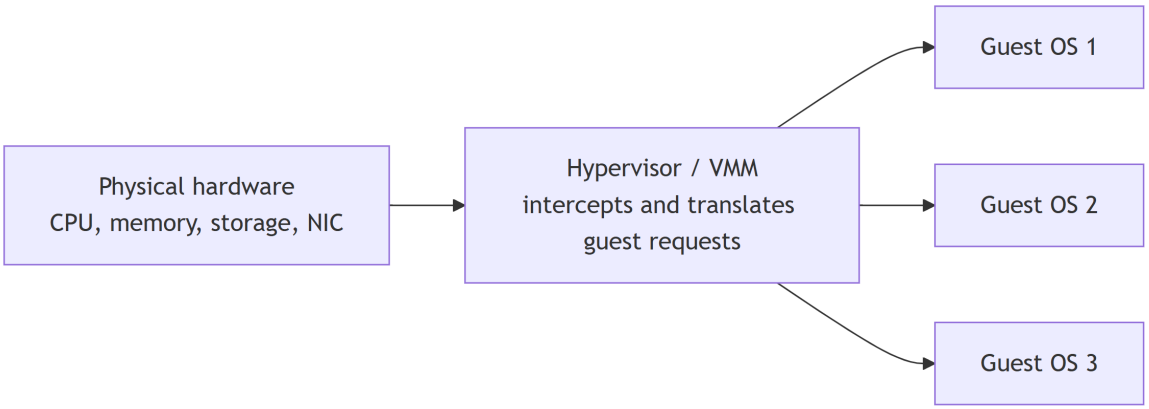
1.4. Hypervisor

Short description: A hypervisor, or Virtual Machine Monitor (VMM), is software that lets multiple operating systems run on a single physical machine.

Theory: It manages hardware resources such as CPU, memory and storage and allocates them to virtual machines without interference. This improves hardware utilization, reduces costs, and provides flexibility in cloud and server environments.

How it works: A hypervisor runs on hardware or on a host OS to create and manage virtual machines, each with its own virtual CPU, memory, storage and network. It intercepts guest OS requests and translates them to physical hardware, ensuring isolation, security and stability.

1.4.1 Architecture diagram



1.4.2 Type 1 hypervisor

A Type 1 hypervisor runs directly on the host's hardware and does not rely on a host operating system. This architecture offers better performance and security because there is no intermediary OS. It is the standard for enterprise-level data centres and for cloud providers such as Amazon Web Services and Microsoft Azure.

Examples: VMware ESXi, Microsoft Hyper-V, KVM (Kernel-based Virtual Machine), Xen.

- **Pros:** high performance through direct hardware access; strong security with no intermediate OS layer; suitable for mission-critical workloads.
- **Cons:** requires dedicated hardware; setup and management are complex compared to Type 2.

1.4.3 Type 2 hypervisor

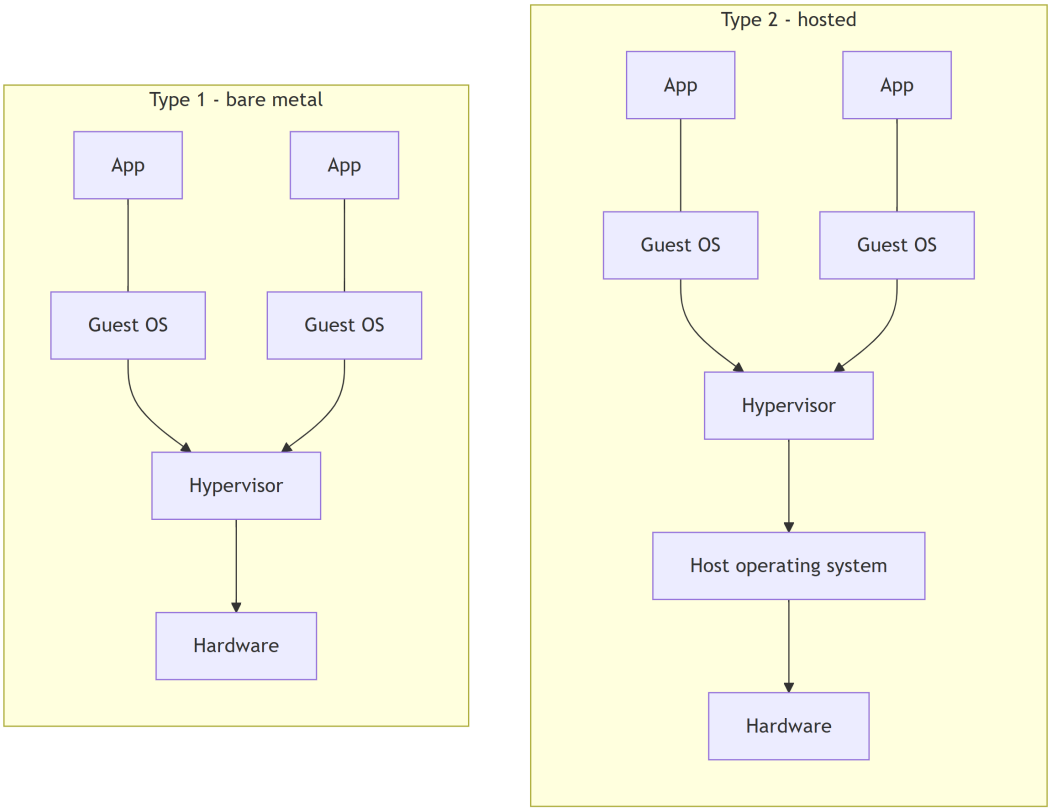
A Type 2 hypervisor runs on top of a conventional operating system such as Windows, macOS or Linux. It is essentially an application within the host OS. This type is generally used for desktop virtualization, development and testing, where a user needs to run multiple operating systems on a personal computer. Performance is slightly lower than Type 1 because of the overhead of the host OS.

Examples: Oracle VM VirtualBox, VMware Workstation, Parallels Desktop.

- **Pros:** easy to install and use; useful for development, testing and malware analysis; good host–guest integration features.

- **Cons:** slower performance with no direct hardware access; security depends on the host OS, so a compromise of the host may affect the guests.

1.4.4 Type 1 versus Type 2



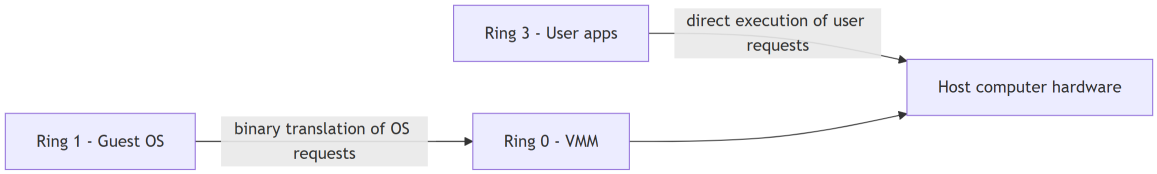
1.5. Full Virtualization

Short description: Full virtualization was introduced by IBM in 1966. It was the first software solution for server virtualization and uses binary translation together with direct execution.

How it works: The virtual machine completely isolates the guest OS from the virtualization layer and the hardware. Kernel code is translated to replace non-virtualizable instructions with new sequences of instructions that have the intended effect on the virtual hardware, while user level code is executed directly on the processor for high performance.

Examples: Microsoft and Parallels systems.

1.5.1 Architecture diagram



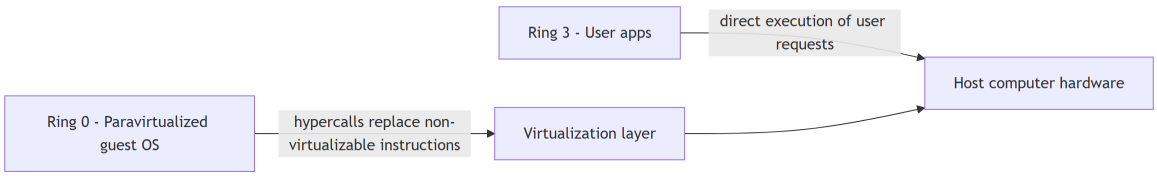
1.6. Paravirtualization

Short description: Paravirtualization is the category of CPU virtualization that uses hypercalls for operations handled at compile time.

How it works: The guest OS is not completely isolated, only partially isolated by the virtual machine from the virtualization layer and hardware. The OS kernel is modified to replace non-virtualizable instructions with hypercalls that communicate directly with the virtualization layer, which improves performance and efficiency.

Examples: VMware and Xen.

1.6.1 Architecture diagram

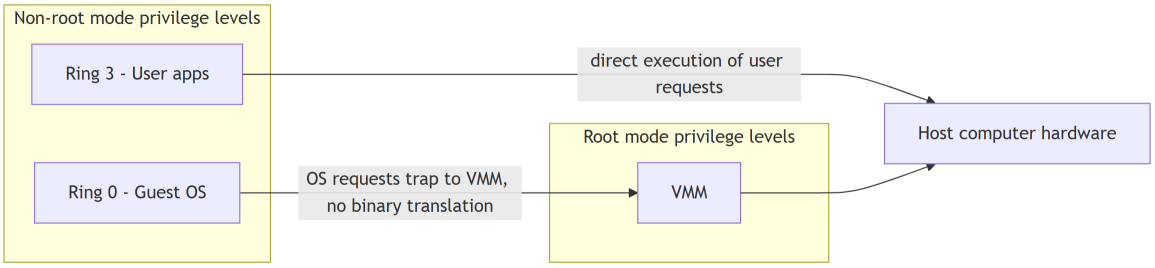


1.7. Hardware Based Virtualization

Short description: Hardware vendors added CPU features that simplify virtualization directly in silicon.

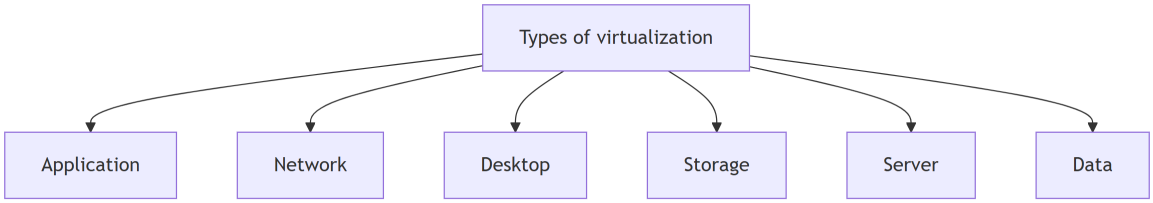
How it works: First generation enhancements include Intel Virtualization Technology (VT-x) and AMD's AMD-V. Both target privileged instructions with a new CPU execution mode that allows the VMM to run in a root mode below ring 0. Privileged and sensitive calls are set to trap automatically to the hypervisor, removing the need for either binary translation or paravirtualization. The guest state is stored in Virtual Machine Control Structures (VT-x) or Virtual Machine Control Blocks (AMD-V).

1.7.1 Architecture diagram



1.8. Types of Virtualization

1.8.1 Overview

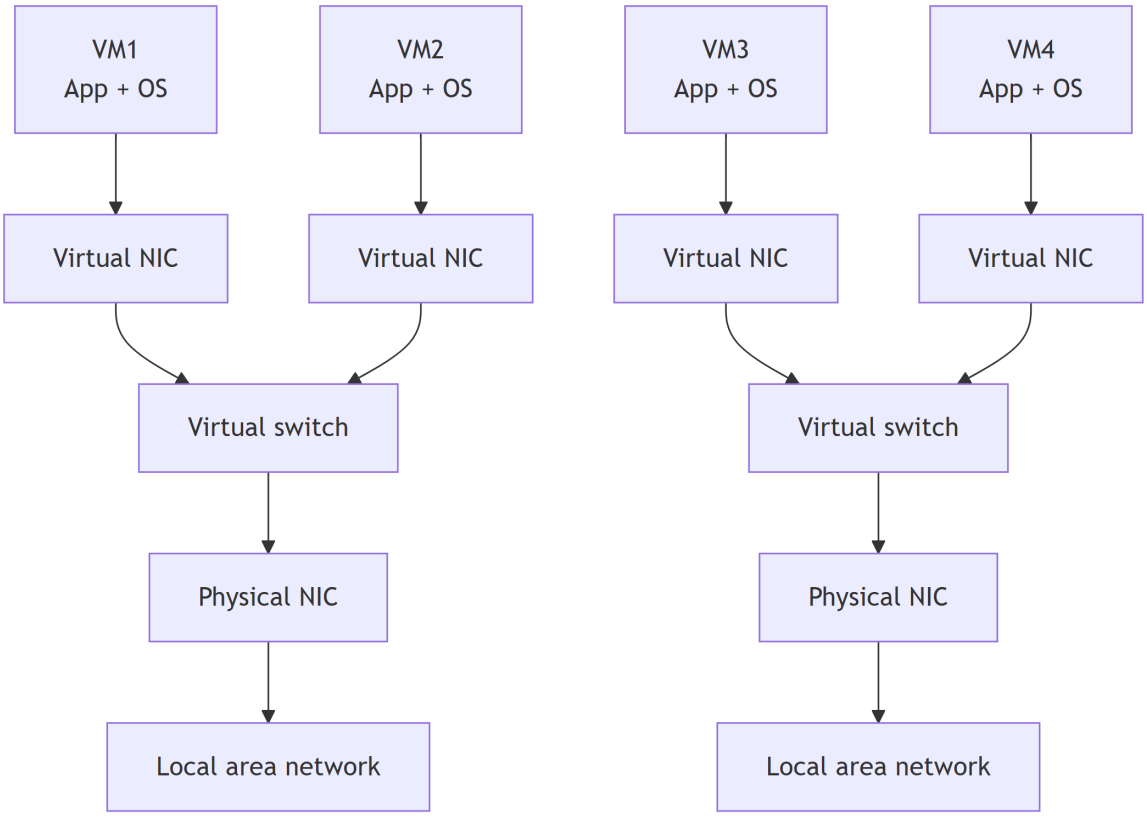


1.8.2 Desktop virtualization

Also called client virtualization. The virtualization happens on the user's site: desktops are replaced with thin clients that use datacentre resources.

1.8.3 Network virtualization

Part of the virtualization infrastructure, used especially when virtualizing servers. It allows multiple switches, VLANs and NAT-ing to be created in software.

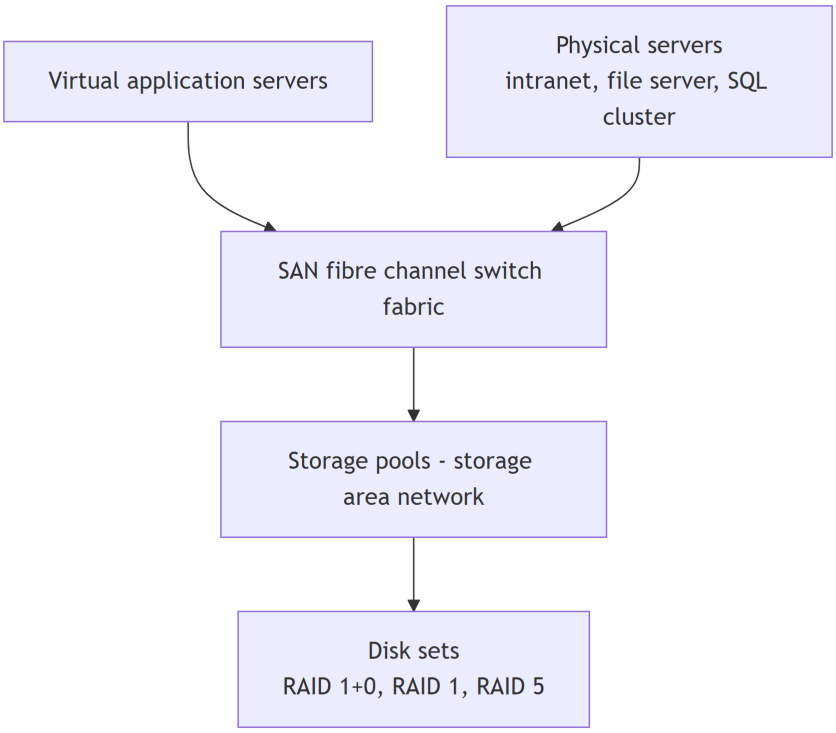


1.8.4 Server and machine virtualization

Virtualizing the server infrastructure, so no further physical servers are needed for different purposes.

1.8.5 Storage virtualization

Widely used in datacentres with large storage. It helps create, delete and allocate storage to different hardware, over a network connection. The leading technology here is the SAN.



1.8.6 Operating system level virtualization

An abstraction layer between the traditional OS and user applications. OS-level virtualization creates isolated **containers** on a single physical server, using one OS instance to utilize the hardware and software in data centres. The containers behave like real servers. It is commonly used to create virtual hosting environments that allocate hardware resources among a large number of mutually distrusting users, and to a lesser extent to consolidate server hardware by moving services on separate hosts into containers or VMs on one server.

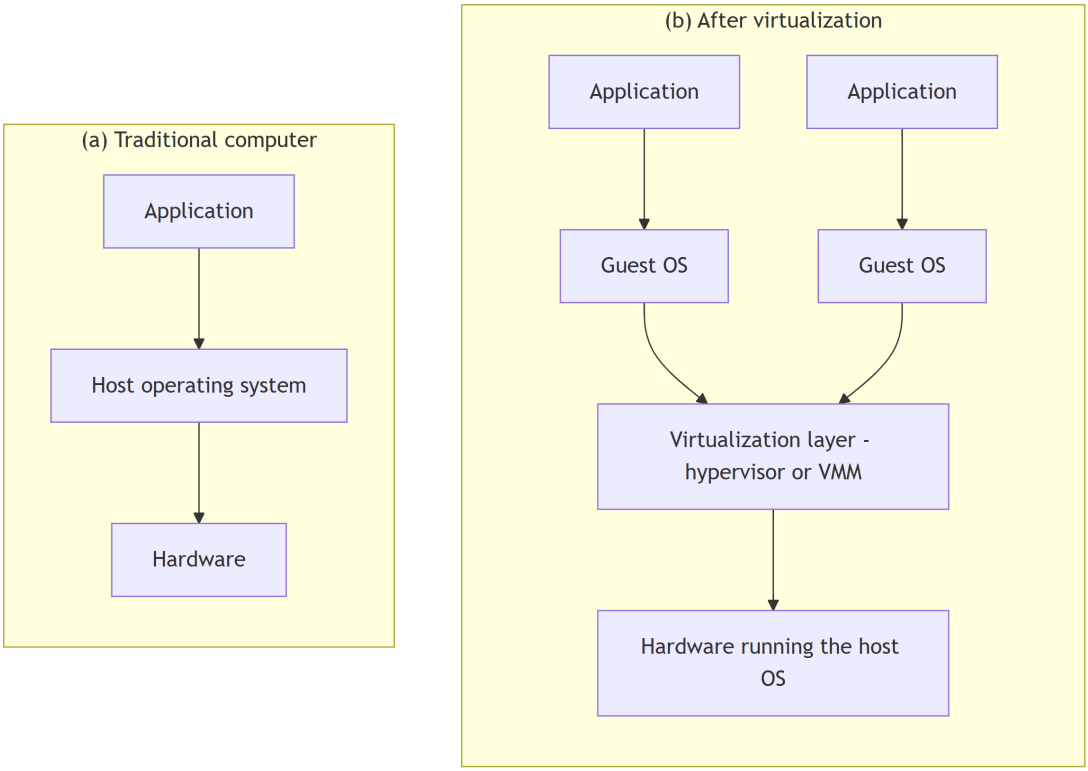
1.8.7 Application virtualization

The technology isolates applications from the underlying operating system and from other applications, to increase compatibility and manageability. Docker can be used for this purpose.

1.9. Levels of Virtualization

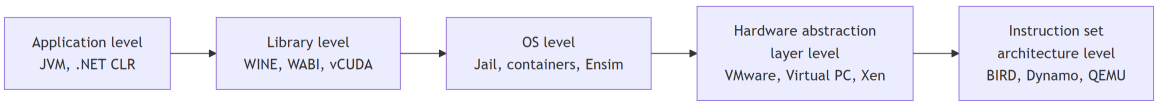
Short description: A traditional computer runs a host operating system tailored to its hardware architecture. After virtualization, different user applications managed by their own guest operating systems can run on the same hardware, independent of the host OS. This is done by adding a software layer called the virtualization layer, known as the hypervisor or virtual machine monitor.

1.9.1 Before and after virtualization



1.9.2 Implementation levels

Virtualization can be implemented at different levels within a computing system, offering varying degrees of isolation and flexibility.



1.9.3 Instruction set architecture level

- Virtualization is performed by emulating a given ISA with the ISA of the host machine.
- For example, MIPS binary code can run on an x86-based host machine through ISA emulation.
- It becomes possible to run a large amount of legacy binary code written for various processors on any given new hardware host.

1.9.4 Hardware abstraction level

- Hardware-level virtualization is performed right on top of the bare hardware. It generates a virtual hardware environment for a VM, and the process manages the underlying hardware through virtualization.
- The idea is to virtualize a computer's resources, such as its processors, memory and I/O devices, to upgrade the hardware utilization rate by multiple concurrent users.
- It was implemented in the IBM VM/370 in the 1960s. More recently the Xen hypervisor has been applied to virtualize x86-based machines running Linux or other guest OS applications.

1.9.5 Operating system level

- An abstraction layer between the traditional OS and user applications.
- It creates isolated containers on a single physical server, with the OS instances utilizing the hardware and software in data centres.
- The containers behave like real servers, and are commonly used for virtual hosting environments allocating hardware among mutually distrusting users.

1.9.6 Library support level

- Most applications use APIs exported by user-level libraries rather than lengthy system calls, so a well-documented API becomes another candidate for virtualization.
- Virtualization with library interfaces controls the communication link between applications and the rest of the system through API hooks.
- WINE implements this approach to support Windows applications on top of UNIX hosts.
- vCUDA allows applications executing within VMs to leverage GPU hardware acceleration.

1.9.7 User-application level

- Virtualization at the application level virtualizes an application as a VM. On a traditional OS an application runs as a process, so this is also known as process-level virtualization.
- The most popular approach is to deploy high level language (HLL) VMs. The virtualization layer sits as an application program on top of the operating system and exports the abstraction of a VM that runs programs compiled to a particular abstract machine definition.
- The Microsoft .NET CLR and the Java Virtual Machine are two good examples.

1.10. Containers

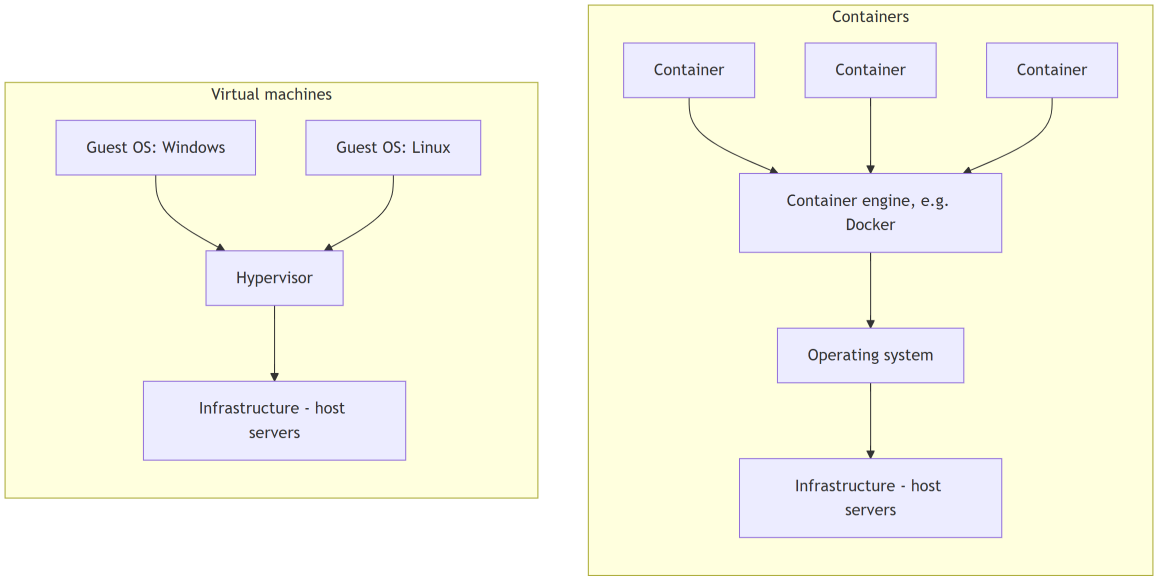
Short description: A container is an isolated, stand-alone unit that encapsulates an application and all its dependencies. It runs consistently in any environment, independent of the host system.

Theory: It is a light, executable software package that wraps everything an application needs: libraries, configuration files and binaries. The application behaves the same way on a developer's laptop, in a testing environment, or on a production server. Unlike traditional virtual machines that carry a full OS, containers pack only what is required, which makes them faster and more efficient.

How it works: Containerization is the process of packing an application together with all its dependencies into a container so it runs consistently from one computing environment to another. In simple terms it uses the host OS kernel to run many isolated instances of applications on the same machine, which makes it very lightweight and efficient.

1.10.1 Containers versus virtual machines

Both containers and VMs run applications in isolated environments, but they differ in architecture, performance, resource usage and startup time.



1.10.2 Comparison

	Containers	**Virtual machines**
Architecture	All containers share the host OS kernel; the running user spaces are isolated, which makes them lightweight.	A hypervisor running on the host OS includes a full guest OS with virtualized hardware.
Boot time	Much less, typically seconds, as no full OS has to boot.	Longer, because the full OS in the VM must be initialized.
Isolation	At the process level, less strong than a VM, though for many use cases this does not matter.	Very good, because each VM is a system on its own with its own OS.
Resource usage	Fewer resources: only the necessary binaries and libraries, not an entire OS.	Very high, as the full OS overhead is incurred for each instance.

1.11. Container Orchestration

Short description: Container orchestration is the process of automating the deployment, management, scaling and networking of containers throughout their lifecycle.

Theory: Orchestration platforms such as Kubernetes create a smooth, self-managing operation and coordinate microservices held in separate containers. Kubernetes eliminates many of the manual processes involved in deploying and scaling containerized applications, and Red Hat creates essential tools for securing, simplifying and automatically updating container infrastructure.

1.11.1 Why it is needed

When many containers run across different machines, managing them manually becomes difficult. An orchestration platform automates:

- **Deployment:** starts and distributes containers across servers.
- **Scaling:** increases or decreases the number of container instances based on demand.
- **Load balancing:** distributes incoming traffic among healthy containers.
- **Self-healing:** restarts failed containers or moves them to healthy nodes.
- **Service discovery:** allows containers to find and communicate with each other.
- **Rolling updates and rollbacks:** updates applications with minimal downtime and reverts changes if needed.
- **Resource management:** efficiently allocates CPU, memory and storage.

Benefits

Orchestration platforms automate the entire lifecycle of containers, transforming what would be a non-stop maintenance operation for IT teams into a smooth, self-managing one, and bring advantages in development methods, costs and security.

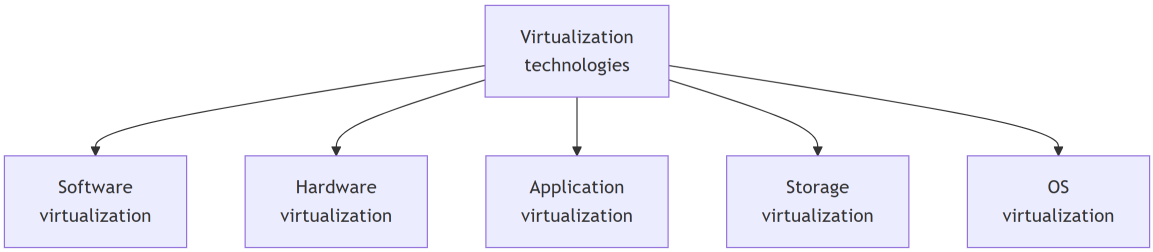
Unit 2: Virtualization Technologies and Data Center Management

2.1. Virtualization Technologies

Short description: Unit 1 covered what virtualization is and how a hypervisor works. This unit covers the technologies that implement it, the data centre those technologies run in, and the addressing scheme that ties the machines together.

Five technologies are studied in this unit.

1. Software virtualization
2. Hardware virtualization
3. Application virtualization
4. Storage virtualization
5. Operating system virtualization



2.2. Software Virtualization

Short description: Software virtualization is a technique that allows a server to run multiple operating systems and applications by creating a virtual environment.

Theory: Software virtualization develops a virtual environment in which software, applications and operating systems can be installed. A virtualized software is simply an application that has been installed inside such an environment. It lets a user run more than one operating system on the same machine. VMware and VirtualBox are common examples.

2.2.1 Components

- **Hypervisor:** a piece of software that manages and allocates hardware resources to the virtual machines.
- **Virtual machines:** independent computers, each running its own operating system on a portion of the physical machine's hardware.
- **Abstraction layer:** the layer of separation between the physical hardware and the virtual machines.

2.2.2 Types of software virtualization

The three forms studied in this unit are operating system virtualization, application virtualization and service virtualization. Each is covered in its own section below.

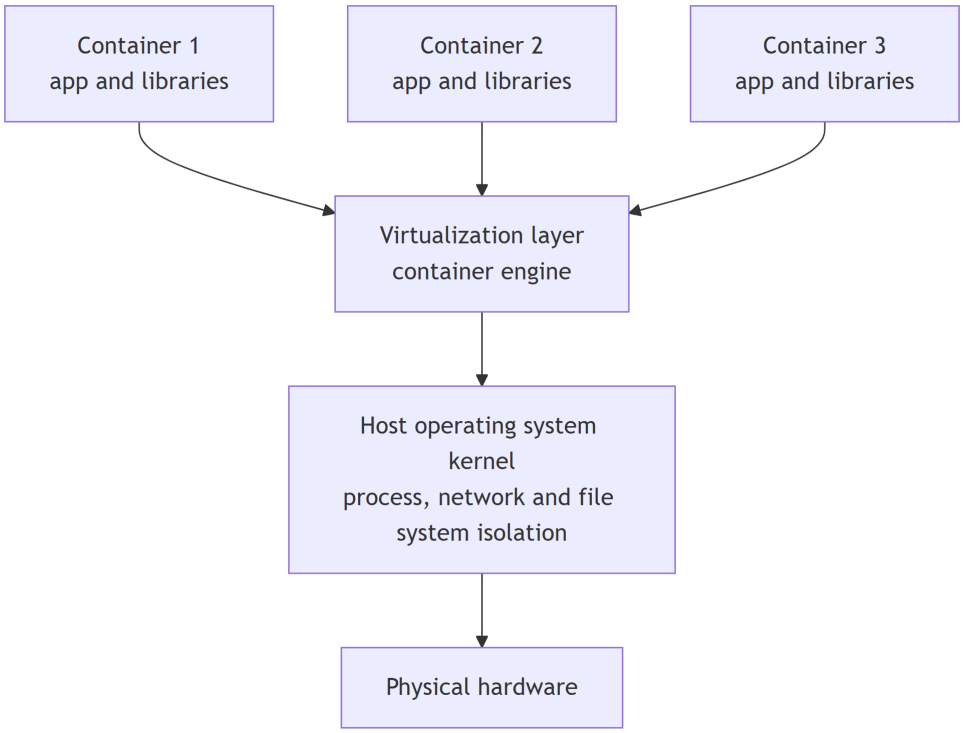
2.3. Operating System Virtualization

Short description: A virtualization technique in which the virtualization software runs on top of a host operating system and creates isolated environments called containers.

Theory: Multiple applications or services run independently on the same physical hardware while sharing the underlying OS kernel. The virtualization layer creates an abstraction between the hardware and the applications, which improves resource utilization and isolation. Unlike traditional virtualization, which creates complete virtual machines, OS-based virtualization shares the host kernel among the containers, so it is lighter and more efficient.

How it works: The host operating system kernel manages the containers through specialised features that control resource access and allocation. Each container operates as an isolated environment with its own process space, network interfaces and file systems, while sharing the underlying kernel. The kernel enforces limits on CPU time, memory usage, disk space and network bandwidth for each container, so one container cannot affect the others and no conflicts or security breaches arise between them.

2.3.1 Architecture diagram



2.3.2 Types of virtual disk

- **Private virtual disk:** dedicated to a single container. It keeps data and configuration across restarts, much like a local hard disk.
- **Shared virtual disk:** used by several containers at the same time. Changes go to a temporary cache that is cleared on restart, which gives every container the same baseline configuration.

2.3.3 Advantages

- **Resource efficiency:** containers share the host kernel, so memory and storage overhead is lower than with full virtualization.
- **Fast startup:** a container starts quickly because no full OS boot is needed.
- **High density:** more containers fit on the same hardware than virtual machines.
- **Portability:** containers move easily between host systems.

2.3.4 Disadvantages

- **OS dependency:** every container must use the same operating system as the host.
- **Security concerns:** a kernel vulnerability can affect all containers at once.
- **Limited isolation:** isolation is weaker than in hardware-based virtualization.

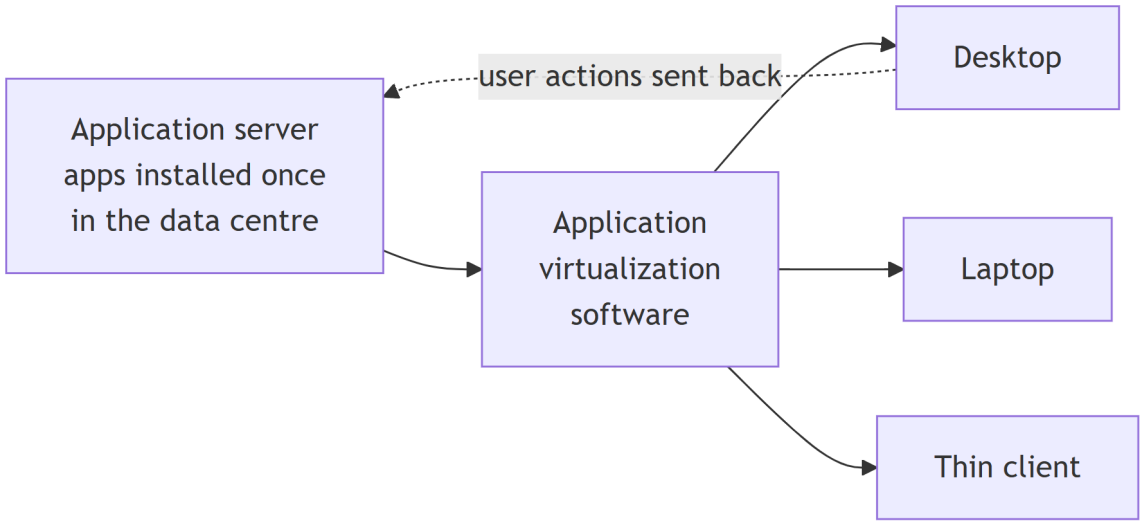
2.4. Application Virtualization

Short description: Application virtualization is the ability to run an application in an environment different from the one it was developed for.

Theory: It allows IT administrators to install applications on a server and then deliver them remotely to users' computers. The experience of using a virtualized application is the same as using an application installed on a physical machine.

How it works: The common approach is server based. An administrator installs the applications once, either in the organisation's data centre or through a hosting service, and then uses application virtualization software to deliver them to users' desktops and other connected devices. The user accesses the application as if it were installed locally, and the user's actions are conveyed back to the server to be executed. Application virtualization is an integral part of digital workspaces and desktop virtualization.

2.4.1 Architecture diagram



2.4.2 Benefits

- **Simplified management:** the applications to be delivered are installed on a single server rather than on every user device, which makes installing software, updates and patches far easier for the IT team.
- **Scalability:** administrators can deploy virtual applications to every connected device regardless of its operating system or storage capacity. The organisation then spends less on hardware, because employees only need a basic computer to reach the applications they work with.
- **Security:** access to applications is controlled centrally, so licences are easy to revoke when a user no longer needs them. The application never has to be uninstalled from a user's device, because it stays resident but reachable only through its container. This central control matters most when a device is lost or stolen, since sensitive data can no longer be reached remotely.

2.5. Service Virtualization

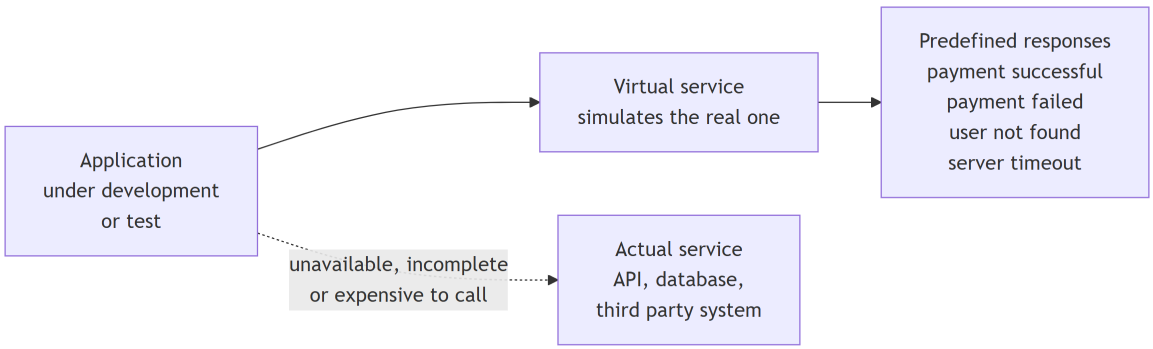
Short description: Service virtualization simulates the behaviour of software services, APIs, databases or third-party applications.

Theory: It lets developers and testers build, integrate and test an application even when the real service is unavailable, incomplete, expensive to access or difficult to configure. In simple terms, it creates a virtual version of a dependent service that behaves like the real one without requiring the actual system.

How it works: Instead of communicating with the actual service, the application talks to the virtual service, which returns predefined or dynamically generated responses.

Examples: Typical simulated responses are payment successful, payment failed, user not found, server timeout and invalid request.

2.5.1 Architecture diagram



2.6. Hardware Virtualization

Short description: Hardware virtualization allows multiple virtual machines to run on a single physical machine.

Theory: It creates a virtual version of a computer's hardware components: the CPU, memory, storage and network. The resulting virtual environment, the virtual machine, acts like a real computer and runs its own operating system and applications independently of the host system.

How it works: A software layer called the hypervisor creates and manages the virtual machines on a single physical server. The hypervisor allocates the server's resources, such as CPU, memory and storage, to each VM, so several operating systems and applications run independently on the same hardware.

2.6.1 Benefits

- **Improved resource utilization:** consolidating several workloads onto one physical server makes better use of the hardware, so fewer physical servers are needed and energy consumption and floor space both drop.
- **Cost savings:** fewer machines means savings on both capital expenditure, which is buying the servers, and operational expenditure, which is maintaining and powering them.
- **Flexibility and scalability:** virtual machines are easy to create, modify and move, which matters for a business that has to adapt quickly to changing workloads.
- **Isolation and security:** each VM operates in isolation, so malware or a system failure is contained within a single VM.
- **Simplified management:** centralised tools let administrators manage many virtual environments from one interface, which is especially useful when the infrastructure is spread across several locations.

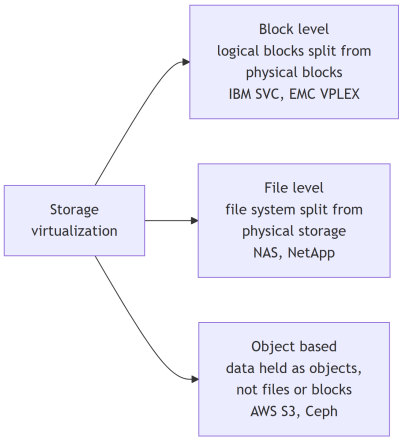
2.7. Storage Virtualization

Short description: Storage virtualization brings several storage devices together into a single entity, for better efficiency, security and management.

Theory: Storage virtualization is a newer concept under virtualization: storage systems use it to gain functionality and features. A storage system is also called a storage array, a disk array or a filer. It uses special hardware and software to provide fast and reliable storage for computing and processing data, and it is designed to accommodate large capacity together with data protection.

How it works: Multiple physical storage devices are combined into a single logical storage pool by a virtualization layer. That layer manages allocation, maps logical volumes to physical devices and handles every read and write transparently. Users and applications see a single logical storage system, while the software manages the physical storage underneath and improves utilization, scalability, performance and ease of management.

2.7.1 Types of storage virtualization



- **Block level virtualization:** separates logical storage from physical storage at the block level, as in IBM SVC and EMC VPLEX.
- **File level virtualization:** abstracts the file system from the physical storage, as in NAS solutions such as NetApp.
- **Object based storage:** holds data as objects rather than as files or blocks, as in AWS S3 and Ceph.

2.7.2 Benefits

- **Enhanced storage utilization:** aggregates different storage devices efficiently.
- **Scalability:** storage scales easily as requirements grow.
- **Savings:** hardware cost falls because existing resources are reused.
- **Very high availability:** redundancy and failover support are built in.
- **Improved performance:** workload is distributed dynamically across the storage.
- **Easier data migration:** data moves from one storage system to another seamlessly.
- **Energy efficiency:** optimised storage usage draws less power.

2.8. The Data Center

Short description: A data centre is a centralised facility that houses computing resources: servers, storage systems, networking equipment, power systems, cooling systems and security infrastructure.

Theory: It is designed to store, process, manage and distribute large amounts of data and applications continuously and securely. In simple terms, the data centre is the backbone of modern IT infrastructure, providing the environment needed to run business applications, websites, cloud services, databases and enterprise systems.

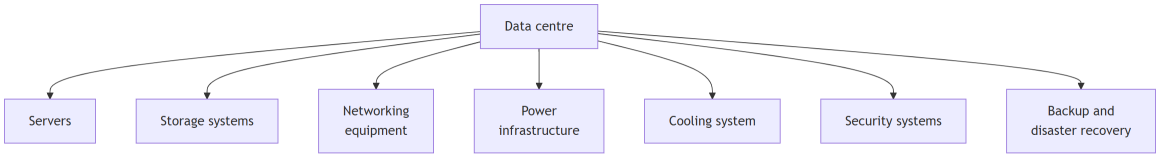
2.8.1 Need for a data centre

- Centralised storage and management of data
- High-speed data processing
- Secure storage of business information
- Continuous availability of services
- Disaster recovery and backup
- Support for cloud computing and virtualization
- Efficient resource utilization

2.8.2 Objectives of a data centre

- Provide reliable IT services
- Ensure high availability of applications
- Store and process large volumes of data
- Protect data against failures and cyber threats
- Improve business continuity
- Optimise resource utilization

2.8.3 Components

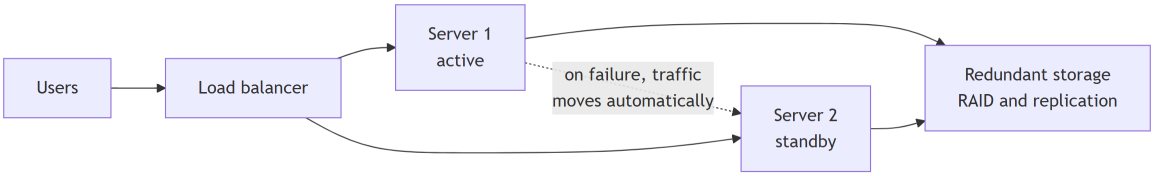


2.9. High Availability of a Data Center

Short description: High availability is the capability of a data centre to keep applications, servers, storage and network services available with minimal or no downtime, even when hardware, software or network failures occur.

Theory: The primary objective is continuous service to users, achieved by eliminating single points of failure and by enabling automatic recovery from failures.

2.9.1 Architecture diagram



2.9.2 Need for high availability

Organisations require uninterrupted service, for the following reasons.

- Continuous business operations
- Better customer satisfaction
- Prevention of financial loss
- Protection against hardware failures
- Data protection
- Disaster recovery
- Compliance with service level agreements
- Improved business reputation

2.9.3 Causes of data centre downtime

Category	Typical causes
Hardware failures	Server crash, hard disk failure, storage controller failure, memory failure, power supply unit failure
Software failures	Operating system crash, application bugs, database corruption, configuration errors
Network failures	Router failure, switch failure, ISP outage, cable damage
Human errors	Wrong configuration, accidental deletion, improper maintenance
Power failures	Power outage, UPS failure, generator malfunction

2.10. Data Center Maintenance

Short description: Data centre maintenance is the set of processes that keep a data centre operating as required, for example to meet an uptime objective such as 99.9 per cent or a service level agreement.

Theory: The tasks are monitoring and inspecting the equipment and systems, from hardware to electrical, cooling and cabling, and then carrying out the maintenance and repair work needed to meet the operational objectives.

2.10.1 Why it matters

Maintenance underpins most of the IT services a business runs. If the business runs its ERP software in its own data centre, maintenance is what keeps the ERP working. Maintenance is also cheaper than the alternative: system failures are stressful and costly to handle, and proactive work detects problems before they cause trouble.

The following problems arise from insufficient or unplanned maintenance.

- **Equipment failures:** neglected cooling leads to hot spots, which cause equipment to overheat and fail.
- **Power outages:** poor maintenance of the power system causes an outage, which disrupts service and can even lose data.
- **Cabling problems:** badly set up and managed cables lead to unintended disconnections.
- **Water damage:** ignoring water lines and temperature changes leads to flooding, which makes equipment fail.
- **Cleaning problems:** accumulated dust and debris interrupt air flow, damage sensitive equipment and can even start fires.

2.11. Planned Maintenance

Short description: Planned maintenance is maintenance scheduled in advance and carried out during predefined maintenance windows, so that disruption to users and business operations is minimal.

Theory: Because the work is planned, users are informed beforehand, backups are taken and contingency plans are prepared.

2.11.1 Objectives

- Improve system performance
- Prevent equipment failure
- Increase system reliability
- Enhance security
- Extend equipment lifespan
- Ensure business continuity

2.11.2 Characteristics

- Scheduled in advance
- Performed during maintenance windows
- Users are notified before the work starts
- Backup systems are prepared
- Low risk of unexpected failure
- Easy to manage and monitor

2.11.3 Activities

Area	Activities
Hardware	Cleaning servers and racks, replacing faulty hard disks, upgrading RAM, replacing power supplies and cooling fans, checking UPS batteries, generator maintenance. For example, replacing an aging hard disk before it fails.
Software	Operating system updates, security patches, application and database upgrades, firmware and driver updates. For example, applying the latest security patches to Linux servers.
Network	Router firmware updates, switch software upgrades, firewall configuration updates, cable inspection, network optimisation. For example, upgrading switch firmware for stability.
Storage	RAID health check, storage expansion, disk replacement, SAN firmware update, backup verification.
Security	Antivirus updates, firewall rule updates, access control review, password policy updates, security auditing.
Backup	Testing backups, verifying recovery procedures, restoring sample data, disaster recovery testing.
Cooling	Cleaning air filters, checking cooling units, temperature monitoring, airflow optimisation.
Power	UPS testing, generator testing, battery replacement, PDU inspection.

2.11.4 Advantages

- Reduces unexpected failures
- Improves system availability
- Enhances security
- Increases equipment life
- Reduces repair costs
- Better resource planning
- Minimises business disruption

2.11.5 Disadvantages

- Requires scheduled downtime, unless redundant systems are in place
- Needs careful planning
- May temporarily affect users
- Requires extra manpower and coordination

2.12. Unplanned Maintenance

Short description: Unplanned maintenance is unexpected maintenance caused by a sudden failure, fault or emergency. It is not scheduled in advance and needs immediate action to restore normal operation and keep downtime short.

2.12.1 Causes

- Hardware failures in servers, storage devices or network switches
- Power outages or UPS failure
- Cooling system malfunction
- Network connectivity issues
- Software crashes or operating system failures
- Cyberattacks such as malware, ransomware and DDoS
- Human errors such as accidental deletion or wrong configuration
- Fire, flood or other natural disasters

2.12.2 Common activities

1. **Emergency hardware replacement:** replace failed hard disks, SSDs, RAM, CPUs, power supplies or network cards, and faulty switches, routers or cables.
2. **Server recovery:** reboot crashed servers, restore failed virtual machines and recover operating systems from backup images.
3. **Storage recovery:** replace failed disks in RAID arrays, rebuild the RAID configuration and restore lost data from backups.
4. **Network troubleshooting:** find and repair broken links, replace damaged switches or routers and reconfigure network settings after a failure.
5. **Power system recovery:** repair or replace faulty UPS batteries, restore generator functionality and switch to backup power during a utility failure.
6. **Cooling system repair:** fix malfunctioning air conditioning units, restore temperature and humidity control and prevent servers from overheating.

2.12.3 Steps followed

1. Detect the problem through the monitoring systems.
2. Analyse and identify the root cause.
3. Isolate the affected equipment or service.
4. Repair or replace the faulty components.
5. Restore systems and services.
6. Test functionality to confirm normal operation.
7. Document the incident and the corrective actions.
8. Implement preventive measures so it does not recur.

2.12.4 Challenges

- Unexpected downtime and loss of productivity
- Risk of data loss
- Increased operational costs
- Service level agreement violations and customer dissatisfaction
- Higher workload for the IT staff

2.12.5 Best practices to reduce it

- Implement continuous monitoring and alerting.
- Perform regular preventive maintenance.
- Maintain redundant hardware and network paths.
- Keep reliable backups and disaster recovery plans.
- Use high availability and failover systems.
- Conduct periodic hardware health checks.
- Apply security updates and patches on time.

2.12.6 Planned versus unplanned maintenance

Planned maintenance	**Unplanned maintenance**
Scheduled in advance	Occurs unexpectedly
Preventive in nature	Corrective or emergency in nature
Minimal downtime	Downtime is often unavoidable
Lower maintenance cost	Higher maintenance cost
Better resource planning	Immediate resource allocation required
Reduces the risk of failures	Responds after failures occur

2.13. Server Migration

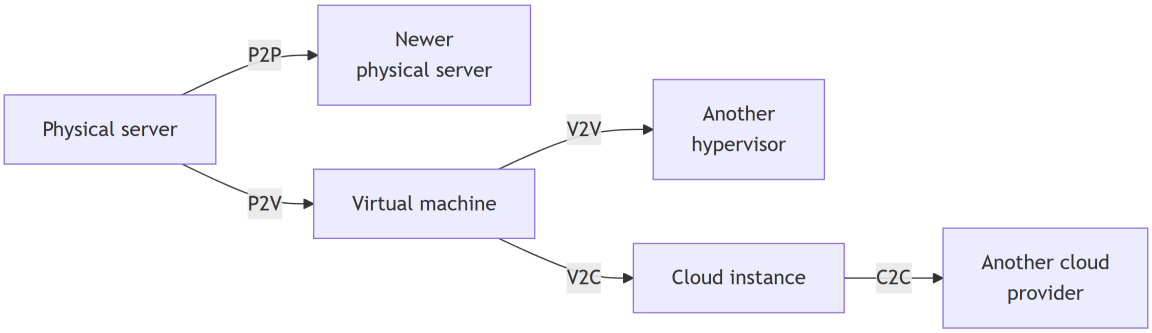
Short description: Server migration is the process of transferring a server's data, processes and configuration to a new target server or cloud instance.

Theory: Organisations migrate servers for resource optimisation, to reduce maintenance overhead, for better support and for deeper integration and modernisation. The process varies with the source and target machine architectures, and it needs careful planning, the right tools and testing to succeed.

2.13.1 Benefits

- **Update to modern services:** move from outdated and restrictive infrastructure to modern, scalable, maintainable platforms. In a cloud environment the servers are virtualized and are called instances.
- **Integration with other services:** modern server environments connect more easily to well-known tools and services, which enables faster automation.
- **Lower management overhead:** migrating to the cloud shifts maintenance from the organisation's own IT team to the hosting provider. On-premises hardware means the internal team handles updates and backups; on cloud infrastructure the provider does.
- **Cost reductions:** on-premises infrastructure is a capital expenditure, while cloud services are billed monthly, annually or pay as you go, which for many organisations is the more efficient budgeting model.
- **Security hardening:** modern platforms carry the latest encryption, identity controls and data security tools, although configuring the environment to suit the organisation's own requirements remains the customer's job.
- **Better performance:** cloud instances reach data held in the same cloud environment faster, and offer a wider range of storage and compute options.

2.13.2 Types of server migration



- **Physical to physical (P2P):** moving a workload from one physical server to another, for example replacing an old server with a newer, faster one.
- **Physical to virtual (P2V):** converting a physical server into a virtual machine. This gives better resource utilization, easier backup, lower hardware cost and simpler management.
- **Virtual to virtual (V2V):** migrating a VM from one virtualization platform to another, for example VMware to Hyper-V or VMware ESXi to KVM.
- **Virtual to cloud (V2C):** moving virtual machines from on-premises infrastructure to a cloud service.
- **Cloud to cloud (C2C):** migrating applications and data between cloud providers, for example AWS to Microsoft Azure or Azure to Google Cloud Platform.

2.13.3 Migration considerations

Business considerations

Migration should align with the business goals. The points to weigh are business continuity, downtime requirements, budget constraints, return on investment, project timeline and regulatory compliance. For example, a hospital migrating its patient databases must keep medical records reachable throughout the migration.

Hardware considerations

Verify that the destination environment meets the hardware requirements: processor architecture, number of CPU cores, RAM capacity, storage capacity, disk type (SSD or HDD) and network interface speed.

Software compatibility

Applications must still work after the move. Check operating system compatibility, database versions, middleware compatibility, drivers and application dependencies.

Data integrity

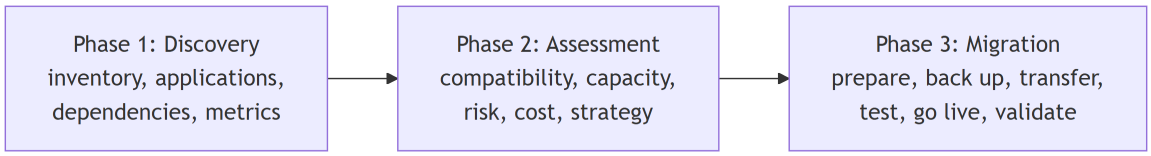
Data must stay accurate and complete throughout the migration. The activities are a full data backup, data validation, checksum verification and database consistency checks.

Self study

Network considerations, security considerations, performance considerations, downtime planning and risk management are left as self study.

2.14. Planning a Server Migration

A successful migration begins with careful planning. The pre-migration process has three phases: discovery, assessment and migration planning.



2.14.1 Phase 1: Discovery

Discovery is the process of collecting complete information about the existing IT infrastructure before the migration. Its purpose is to understand the current environment and identify every resource that has to move.

1. **Server inventory:** server name, IP address, operating system, CPU details, RAM, storage and installed software.
2. **Application discovery:** installed applications, running services, web servers, databases and middleware.
3. **Dependency mapping:** applications rarely work on their own, so identify the dependencies among applications, databases, file servers, authentication servers and external APIs.
4. **Performance monitoring:** CPU utilization, RAM usage, disk usage, network traffic, peak workload and storage growth. These metrics are what size the new environment correctly.
5. **Security discovery:** user accounts, access rights, certificates, firewall rules and encryption methods.

2.14.2 Phase 2: Assessment

Assessment analyses the information collected during discovery, to judge whether the migration is feasible and to prepare a strategy. Its objectives are to evaluate compatibility, estimate resources, identify risks, estimate costs and select the migration method.

1. **Compatibility assessment:** operating system support, application compatibility, database compatibility, driver support and hardware compatibility.
2. **Capacity planning:** estimate future CPU, memory, storage and network bandwidth requirements, so the new environment can carry both the current and the future workload.
3. **Risk assessment:** analyse downtime, performance degradation, security issues, data corruption and hardware failure, and assign a probability and an impact to each.
4. **Cost assessment:** hardware, cloud services, licensing, migration tools, staff training and maintenance.
5. **Migration strategy selection:** covered below.
6. **Backup planning:** take complete backups, verify their integrity, store them securely and prepare rollback procedures.

Migration strategies

Strategy	What it means	Notes
Rehosting	Move applications without changing their architecture, also called lift and shift.	Fast, low cost, minimal modifications.
Replatforming	Make minor optimisations while migrating.	For example, moving a database to a managed cloud database service.
Refactoring	Redesign the application for a cloud-native environment, also called re-architecting.	High scalability, better performance and long-term flexibility, but high cost and complexity.
Repurchasing	Replace the existing software with a SaaS product.	For example, replacing an on-premises email server with Microsoft 365.

2.14.3 Phase 3: Migration

Migration is the actual transfer of applications, operating systems, databases and data to the target environment.

1. **Prepare the target environment:** configure servers, virtual machines, storage, networks and security policies.
2. **Back up the existing systems:** operating system, applications, databases, configuration files and user data.
3. **Transfer the data:** by network transfer, storage replication, backup restoration, migration tools or cloud migration services.
4. **Migrate the applications:** install the applications, restore the databases, configure the services and update the configuration files.
5. **Test:** functional testing, performance testing, security testing and user acceptance testing, to confirm the applications work after the move.
6. **Go live:** redirect user traffic, update DNS records, activate production services and monitor system health.
7. **Post-migration validation:** verify data integrity, application functionality, user authentication, network connectivity, backup operations and performance.

2.14.4 Best practices

- Maintain a complete inventory of servers and applications.
- Perform full backups and verify their integrity.
- Test the migration in a staging environment before production.
- Schedule the migration during off-peak hours to minimise downtime.
- Monitor system performance throughout the migration.
- Prepare and test a rollback plan.
- Validate data integrity and application functionality afterwards.
- Document every step for future reference and audits.

2.14.5 Advantages of proper planning

Reduced downtime, improved application performance, better scalability, efficient resource utilization, lower operational costs, enhanced security, improved disaster recovery and simpler infrastructure management.

2.15. Networking Concepts: Masks and CIDR

Short description: A mask is what separates the network identifier from the host identifier in an IP address.

Theory: A mask is a 32-bit number made of a run of 1s followed by a run of 0s. It does not apply to classes D and E. Classless interdomain routing, or CIDR, writes the mask in the short form `/n`, where `n` is the number of 1 bits.

CIDR	Dotted decimal	Binary	Class
`/8`	`255.0.0.0`	`11111111 00000000 00000000 00000000`	A
`/16`	`255.255.0.0`	`11111111 11111111 00000000 00000000`	B
`/24`	`255.255.255.0`	`11111111 11111111 11111111 00000000`	C

Notes

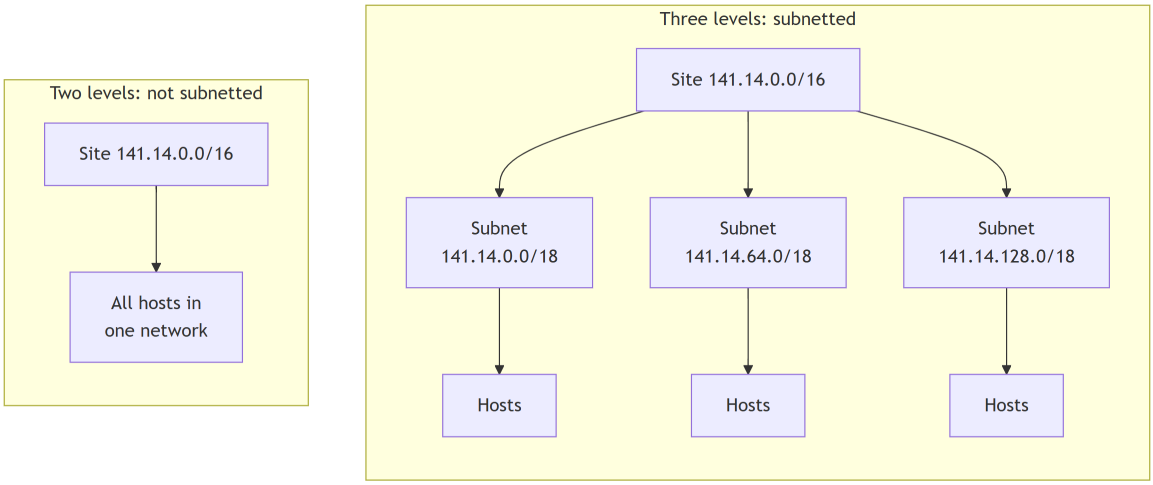
- **Subnetting:** an organisation granted a large block in class A or B divides the addresses into several contiguous groups and assigns each group to a smaller network, called a subnet.
- **Supernetting:** an organisation combines several class C blocks to create a larger range of addresses. Several networks are combined into one supernetwork, or supernet.

2.16. Subnetting

Short description: Subnetting divides a large granted block into several smaller networks, by borrowing bits from the host part and adding them to the network part.

Theory: Subnetting increases the number of 1s in the mask: a subnet mask has more 1s than the corresponding default mask. To build a subnet mask, change some of the leftmost 0s in the default mask to 1s. The number of subnets follows from the number of extra 1s: if there are n extra 1s there are 2^n subnets, and if N subnets are needed the number of extra 1s is $\log_2 N$. The number of subnets must therefore be a power of 2.

2.16.1 Levels of hierarchy



Without subnetting the address has two levels of hierarchy, network and host. With subnetting it has three: network, subnet and host.

2.16.2 Finding the subnetwork address

The subnet address is found the same way as the network address, by applying the mask to the address. There are two ways to do it.

- **Straight method:** write both the address and the mask in binary and apply the AND operation.
- **Short-cut method:** if the byte in the mask is 255, copy the byte of the address. If the byte in the mask is 0, replace the byte of the address with 0. If the byte is neither 255 nor 0, write that byte of the mask and of the address in binary and AND them.

Example: straight method

What is the subnetwork address if the destination address is `200.45.34.56` and the subnet mask is `255.255.240.0`?

Line	Binary
Address	`11001000 00101101 00100010 00111000`
Mask	`11111111 11111111 11110000 00000000`
Result	`11001000 00101101 00100000 00000000`

The subnetwork address is `200.45.32.0`.

Example: short-cut method

What is the subnetwork address if the destination address is `19.30.84.5` and the mask is `255.255.192.0`? The first two bytes of the mask are 255, so `19` and `30` are copied. The last byte of the mask is 0, so `5` becomes `0`. The third byte is neither, so AND it: `01010100` AND `11000000` is `01000000`, which is 64. The subnetwork address is `19.30.64.0`.

Example: designing subnets in class C

A company is granted the site address `201.70.64.0`, a class C address, and needs six subnets.

The default mask of class C has 24 ones. Six is not a power of 2, and the next power of 2 is $8 = 2^3$, so three more 1s are needed. The subnet mask has $24 + 3 = 27$ ones and $32 - 27 = 5$ zeros, which is `11111111 11111111 11111111 11100000`, or `255.255.255.224`. There are 8 subnets and each holds $2^5 = 32$ addresses.

Example: designing subnets in class B

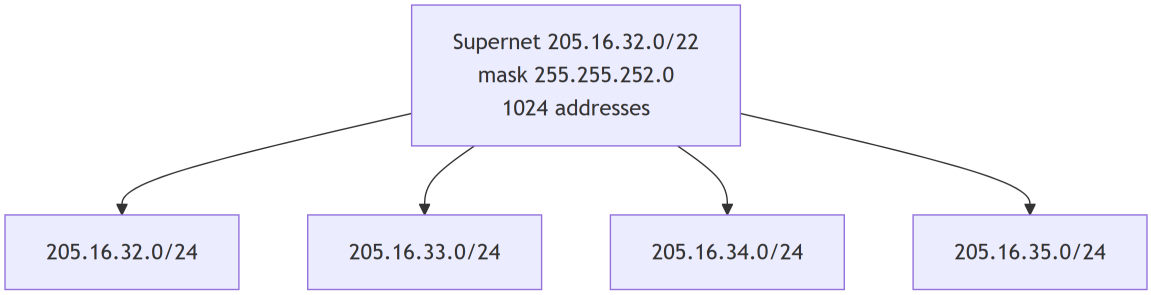
A company is granted the site address `181.56.0.0`, a class B address, and needs 1000 subnets.

The default mask of class B has 16 ones. One thousand is not a power of 2, and the next power of 2 is $1024 = 2^{10}$, so ten more 1s are needed. The subnet mask has $16 + 10 = 26$ ones and $32 - 26 = 6$ zeros, which is `11111111 11111111 11111111 11000000`, or `255.255.255.192`. There are 1024 subnets and each holds $2^6 = 64$ addresses.

2.17. Supernetting

Short description: Supernetting is the opposite of subnetting. Bits are borrowed from the network side to combine a group of networks into one large supernetwork.

Theory: Demand for midsize blocks is high. Class A and class B addresses are almost depleted, while class C addresses are still available but a block of 256 addresses is too small for many organisations. In supernetting, several class C blocks are combined into a larger range. This decreases the number of 1s in the mask: for four class C blocks the mask changes from `/24` to `/22`. An organisation needing 1000 addresses can be granted four contiguous class C blocks forming one supernetwork.



In subnetting, the first address of the subnet and the subnet mask define the range of addresses. In supernetting, the first address of the supernet and the supernet mask define the range.

Example: which packet belongs to the supernet

A supernet has the first address `205.16.32.0` and the supernet mask `255.255.248.0`. A router receives three packets, for `205.16.37.44`, `205.16.42.56` and `205.17.33.76`.

Applying the supernet mask gives `205.16.32.0`, `205.16.40.0` and `205.17.32.0` respectively. Only the first address belongs to this supernet.

Example: valid sets of blocks

A company needs 600 addresses. Which of these sets of class C blocks can form a supernet?

1. `198.47.32.0`, `198.47.33.0`, `198.47.34.0`. No, there are only three blocks.
2. `198.47.32.0`, `198.47.42.0`, `198.47.52.0`, `198.47.62.0`. No, the blocks are not contiguous.
3. `198.47.31.0`, `198.47.32.0`, `198.47.33.0`, `198.47.52.0`. No, 31 in the first block is not divisible by 4.
4. `198.47.32.0`, `198.47.33.0`, `198.47.34.0`, `198.47.35.0`. Yes, all three requirements are fulfilled.

Example: the supernet mask for 16 blocks

Sixteen blocks need four 1s of the default mask changed to 0s, so the mask is `11111111 11111111 11110000 00000000`, or `255.255.240.0`.

Example: size and range of a supernet

A supernet has the first address `205.16.32.0` and the supernet mask `255.255.248.0`. The mask has 21 ones and the default mask has 24, so the difference of 3 gives $2^3 = 8$ blocks. They run from `205.16.32.0` to `205.16.39.0`, so the first address is `205.16.32.0` and the last is `205.16.39.255`.

2.18. Classless Addressing

Short description: Classless addressing removes the classes altogether. Addresses are still granted in blocks, but the size of a block varies with the nature and size of the organisation.

Theory: Classless addressing was designed to overcome address depletion and give more organisations access to the Internet. A household may be given only two addresses, while a large organisation may be given thousands. It is referred to as classless interdomain routing, or CIDR, and the block granted is called a CIDR block.

2.18.1 Rules for allocating a block

1. The IP addresses in the CIDR block must all be contiguous.
2. The block size must be a power of 2, and the size of the block equals the number of IP addresses in it.
3. The first IP address of the block must be divisible by the block size.

For the beginning address, if the block has fewer than 256 addresses only the rightmost byte has to be checked; if it has fewer than 65,536 addresses only the two rightmost bytes have to be checked, and so on.

2.18.2 Slash notation

Slash notation, also called CIDR notation, is a shorter way of writing a subnet mask: instead of the full mask, only the number of 1 bits is written after a slash. A block in classes A, B and C is written `A.B.C.D/n`, where n is 8, 16 or 24 respectively.

2.18.3 Extracting information from an address

Given any address in the block and the prefix length n, three facts follow.

1. The number of addresses in the block is $N = 2^{32-n}$.
2. The first address is found by keeping the n leftmost bits and setting the $32 - n$ rightmost bits to 0.
3. The last address is found by keeping the n leftmost bits and setting the $32 - n$ rightmost bits to 1.

2.18.4 Address mask

The address mask is a 32-bit number whose n leftmost bits are 1 and whose remaining $32 - n$ bits are 0. A computer finds it easily because it is the complement of $2^{32-n} - 1$. Defined this way, the mask lets a program extract the block information with the bitwise NOT, AND and OR operations.

1. Number of addresses: N = NOT(mask) +1.
2. First address: (any address in the block) AND (mask).
3. Last address: (any address in the block) OR NOT(mask).

Example: range of a block

A small organisation is given the block `205.16.37.24/29`. To find the last address, keep the first 29 bits and change the last 3 bits to 1s.

Line	Binary
Beginning	`11001111 00010000 00100101 00011000`
Ending	`11001111 00010000 00100101 00011111`

There are 8 addresses in the block. The same answer follows from the suffix length: it is $32 - 29 = 3$, so there are $2^3 = 8$ addresses, and if the first address is `205.16.37.24` the last is `205.16.37.31`, because $24 + 7 = 31$.

Example: finding the network address

What is the network address if one of the addresses is `167.199.170.82/27`?

The prefix length is 27, so keep the first 27 bits and change the remaining 5 to 0s. Those 5 bits fall in the last byte, which is `01010010`. Changing the last 5 bits to 0 gives `01000000`, or 64. The network address is `167.199.170.64/27`.

Example: subnetting a classless block

An organisation is granted `130.34.12.64/26` and needs four subnets.

The suffix length is 6, so the block holds $2^6 = 64$ addresses. Split four ways, each subnet has 16 addresses. Four subnets need two more 1s on the site prefix, so the subnet prefix is `/28`.

Subnet	Range
1	`130.34.12.64/28` to `130.34.12.79/28`
2	`130.34.12.80/28` to `130.34.12.95/28`
3	`130.34.12.96/28` to `130.34.12.111/28`
4	`130.34.12.112/28` to `130.34.12.127/28`

Example: first and last address of a block

One address of a block is `205.16.37.39/28`. In binary it is `11001101 00010000 00100101 00100111`.

Setting the $32 - 28 = 4$ rightmost bits to 0 gives `11001101 00010000 00100101 00100000`, which is `205.16.37.32`. Setting the same 4 bits to 1 gives `11001101 00010000 00100101 00101111`, which is `205.16.37.47`.

The same result comes from the mask, which for `/28` is `11111111 11111111 11111111 11110000`. ANDing the address with the mask gives the first address. ORing it with the complement of the mask gives the last address. Complementing the mask, reading it as a decimal number and adding 1 gives the number of addresses.

Example: an ISP distributing a block

An ISP is granted `190.100.0.0/16` and has to serve three groups of customers: 64 customers needing 256 addresses each, 128 customers needing 128 addresses each, and 128 customers needing 64 addresses each.

Group	Suffix	Prefix	Sub-blocks
1	8	`/24`	`190.100.0.0/24` to `190.100.63.255/24`, total $64 \times 256 = 16{,}384$
2	7	`/25`	`190.100.64.0/25` to `190.100.127.255/25`, total $128 \times 128 = 16{,}384$
3	6	`/26`	`190.100.128.0/26` to `190.100.159.255/26`, total $128 \times 64 = 8{,}192$

The block granted holds 65,536 addresses and 40,960 of them are allocated, so 24,576 addresses are still available.

2.19. Practice Problems

1. A router receives a packet with the destination address `190.240.33.91`. Show how it finds the network and the subnetwork address to route the packet, assuming the subnet mask is `/19`.

2. An organisation is granted the block `130.56.0.0/16` and the administrator wants 1024 subnets. Find the number of addresses in each subnet, the subnet prefix, the first and last address of the first subnet, and the first and last address of the last subnet.

3. Each of the following addresses belongs to a block. Find the first and the last address in each block: `14.12.72.8/24`, `200.107.16.17/18` and `70.110.19.17/16`.

4. An organisation is granted a block beginning at `14.24.74.0/24` and needs three sub-blocks for its three subnets: one of 10 addresses, one of 60 addresses and one of 120 addresses. Design the sub-blocks.

5. An ISP is granted the block `16.12.64.0/20` and has to allocate addresses for 8 organisations, each with 256 addresses. Find the number and range of addresses in the ISP block, the range for each organisation and the range that stays unallocated, then show the outline of the distribution.

6. An ISP is granted the block `80.70.56.0/21` and has to allocate addresses for two organisations of 500 addresses each, two of 250 addresses each and three of 50 addresses each. Answer the same three parts as the previous problem.